

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

MAY EXAMINATION

UCCD2303 DATABASE TECHNOLOGY /
UCCD2203 DATABASE SYSTEMS

SATURDAY, 11 MAY 2024

TIME : 6.00 PM – 8.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY

Instruction to Candidates:

This question paper consists of **Section A** and **Section B**.

Section A consists of **THREE (3)** questions. Candidates are required to **answer ALL** the questions.

Section B consists of **TWO (2)** questions. Candidates are required to **answer only ONE (1)** question. Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Candidates are not allowed to use a calculator.

Answer questions only in the answer booklet provided.

UCCD2303 DATABASE TECHNOLOGY / UCCD2203 DATABASE SYSTEMS**Section A: Answer ALL QUESTIONS. [Total: 75 Marks]**

- Q1. (a) A DATABASE VIEW is the dynamic result of one or more relational operations operating on base relation(s) to produce another relation. Although there are many benefits in using database view compared to base relation, explain **ANY THREE (3)** disadvantages of database view. (6 marks)
- (b) Specialization hierarchy reflects the one-to-one relationship between the supertype entities and its child entities. Inheritance is an important characteristic to Enhanced Entity Relationship Modelling. Relate **INHERITANCE** to **SPECIALIZATION** in the specialization hierarchy. (6 marks)
- (c) Data might be inconsistent because of errors or duplication in data entry. Therefore, integrity constraints are important to restrict the data values which can be entered into a column in a database table. Differentiate between **ENTITY INTEGRITY** and **REFERENTIAL INTEGRITY**. (6 marks)
- (d) Consider the following scenario:
"One CAR has exactly four WHEELS. However, one WHEEL belongs to exactly one CAR."

Decide whether **AGGREGATION**, **COMPOSITION** or **GENERALIZATION** exists in the given scenario above. Justify your answer(s). (7 marks)
- [Total : 25 marks]

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- Q2. (a) Consider transaction T_1 and T_2 in Table 2.1 and answer the following questions.

Table 2.1 Transaction T_1 and T_2 locking process.

Time	T_1	T_2
t1	Exclusive lock(item A)	
t2	Read(item A)	
t3	Write(item A)	Exclusive lock (B)
t4		Read(item B)
t5		Write(item B)
t6	Exclusive lock (item B)	Exclusive lock (item A)
t7	Read(item B)	Read(item A)
t8	Write(item B)	Write(item A)

- (i) Based on the Table 2.1, draw **ONE (1)** Wait-For-Graph. (3 marks)
 - (ii) Determine the existence of deadlock. Justify your answers. (2 marks)
 - (iii) Provide **ANY FOUR (4)** types of database failure which could occur in the transactions as shown in Table 2.1. (8 marks)
 - (iv) Refer to your answer in Q2(a)(iii), suggest **ANY TWO (2)** operations which may be executed to reach a new consistent state of the transactions. (4 marks)
- (b) ACID stands for Atomicity, Consistency, Isolation and Durability. In order to ensure that all the operations are executed successfully upon completion, the database transactions should be compliant to the so-called ACID properties. Illustrate the **FOUR (4)** basic properties of a transaction with an example for each in a banking environment. (8 marks)
- [Total : 25 marks]

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Q3. Answer all the questions below based on the given scenario in Figure 3.1.

XYXCompany Order Form				
Order ID	20230727-98765			
Order date	27-Jul-2023			
Client name	Sandra Woods			
Client address	123, Jalan Satu, Taman Dua, 30000 Kuala Lumpur.			
Client phone	603-1234567			
Staff name	Jane			
Staff ID	S0001			
Product ID	Product Description	Unit Price (RM)	Quantity	Total (RM)
P2345	Mouse	30.00	10	300.00
P5432	Speaker	20.00	2	40.00
P7890	Microphone	15.00	4	60.00
P4321	HDMI cable	50.00	1	50.00
			Sub-total	450.00
			Sales tax	10.00
			Total	460.00

Figure 3.1. XYXCompany Order Form

- Propose the required ENTITIES in Figure 3.1. (4 marks)
 - Propose the required ATTRIBUTES for each entity and clearly indicate which are the primary key(s) and foreign key(s) for each entity. (5 marks)
 - Propose all the required primary key(s) and foreign key(s) for each of the entity identified in the given scenario above. (5 marks)
 - Identify the CARDINALITIES among the entities based on the answers in Q3(b). (4 marks)
 - Construct **ONE (1)** Entity Relationship Diagram (ERD) for the scenario in Figure 3.1. (7 marks)
- [Total : 25 marks]

UCCD2303 DATABASE TECHNOLOGY / UCCD2203 DATABASE SYSTEMS**Section B Answer ONLY ONE (1) question. [Total: 25 marks]**

Q4. (a) ABCAdvisory is a company which provides audit, consulting, financial advisory, risk advisory, tax, and legal services to clients globally. This company operates across the world in more than 1000 locations in the world. In order to support the company internal customers and external customers, the company requires a fast data processing at several sites remotely participating in the data request processing. As a database analyst, you are required to provide some advice to the management of ABCAdvisory.

- (i) Identify **ANY THREE (3)** possible key issues to be considered when designing a distributed database in the given scenario above. (6 marks)
- (ii) In a distributed computing environment, the requests from users are processed by a Database Management System (DBMS) server. The DBMS is typically remote to the client and a Distributed Database Management System (DDBMS) is required. Which type of DDBMS can be chosen? Justify your answers with **ANY FOUR (4)** reasons for your choice. (8 marks)
- (iii) Explain **ANY FIVE (5)** functionalities of Distributed Database Management System (DDBMS) which suits the needs of ABCAdvisory. (5 marks)

- (b) The objective of query optimization is to choose the one that can minimize the usage of resources. The query as shown below will display all Managers who work at a London branch.

```
SELECT *  
FROM branch b,staff s  
WHERE s.branchNo = b.branchNo  
AND (s.position = 'Manager' AND b.city = 'London');
```

Propose **ANY THREE (3)** equivalent RELATIONAL ALGEBRA for the query above. (6 marks)

[Total : 25 marks]

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- Q5. (a) Different database technologies become available that better fit for newer types of use cases. SQL (Structured Query Language) is used for accessing relational databases. NoSQL is defined as "not only SQL" which is a non-relational database that is used to provide more flexibility to the users. Compare SQL and NoSQL in terms of processing cost, structure of data, and scalability. (6 marks)
- (b) Name **ANY THREE (3)** properties of NoSQL. (6 marks)
- (c) Suggest **ANY THREE (3)** NoSQL tools for visualizing Big Data. (3 marks)
- (d) Apache Hadoop is an Apache top-level project being built as an open source software framework for large storage. The two primary components are the Hadoop Distributed File System (HDFS) and the MapReduce parallel processing framework. Elaborate the Hadoop Ecosystem which allow the system to continue operating in case of node failure. (6 marks)
- (e) Workable solutions are the key challenges in wireless information management. The use of mobile communication devices in the development of a Distributed Database Management System (DDBMS) is crucial. Identify **ANY FOUR (4)** challenges that can happen during the implementation of Mobile DBMSs. (4 marks)

[Total : 25 marks]